

LTSpice Notch Quality Report

CROSS COUPLED

Auto-detected swept parameters and quality-factor ($Q = f_0 / BW$ @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 1

Total simulation steps : 88

V(act+,act-)
swept parameters: Cp, Itail

Report contents (per signal column):

- 1. All curves — Bode plot (magnitude + phase)
- 2. Magnitude grouped by each swept parameter (one subplot per fixed value)
- 3. Phase grouped by each swept parameter (one subplot per fixed value)
- 4. Quality-factor table for that column

End of report (extra analyses):

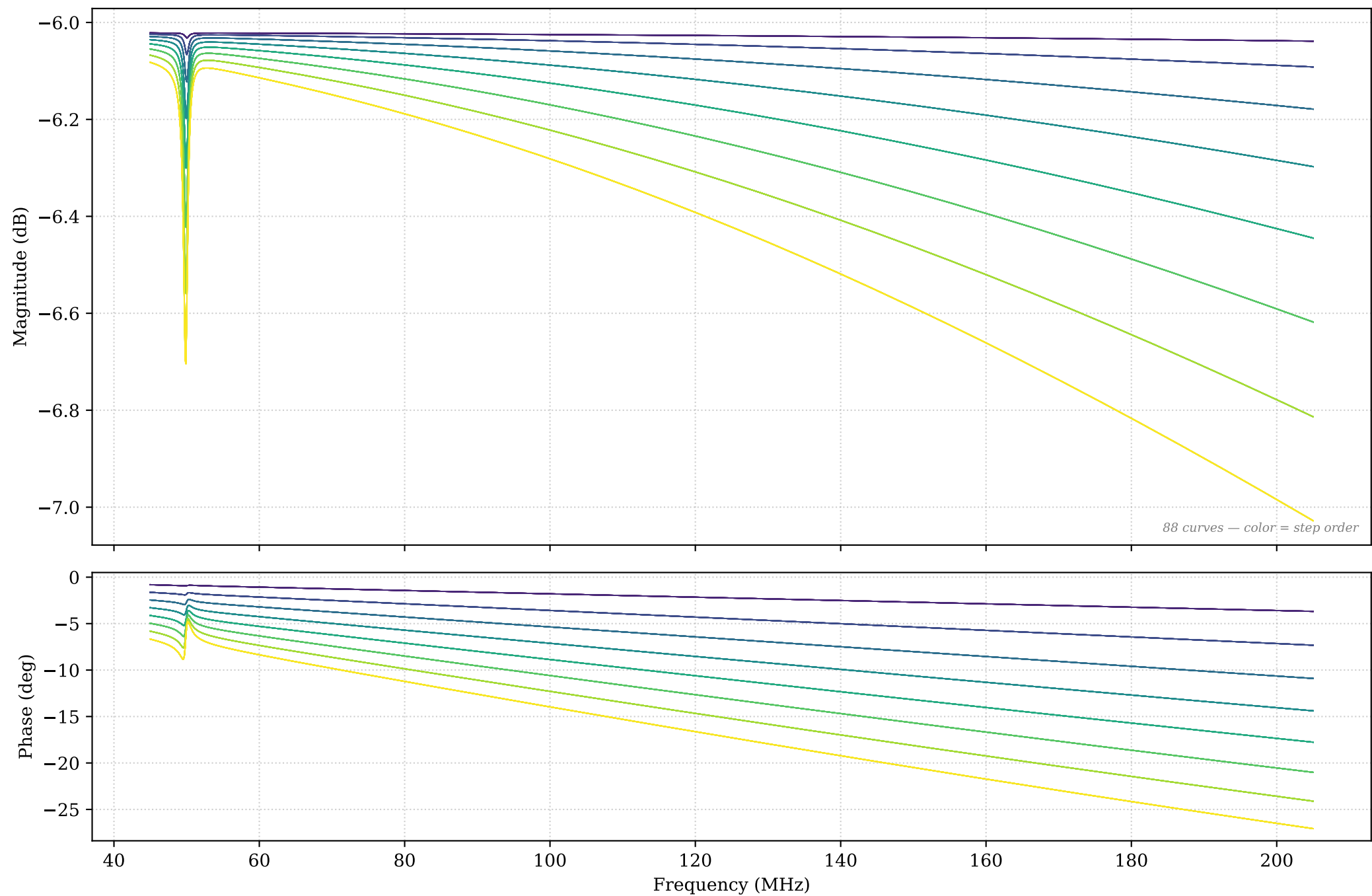
- A. Cross-signal comparison (act vs pas, if both present)
- B. Q vs each swept parameter (one line per other-param combo)
- C. BW vs each swept parameter
- D. Q heatmap for V(act): rows $\times$ columns swept params, colored
- E. Combined quality-factor table across all signals

Signal: $V(\text{act+}, \text{act-})$

Swept parameters: C_p , I_{tail}

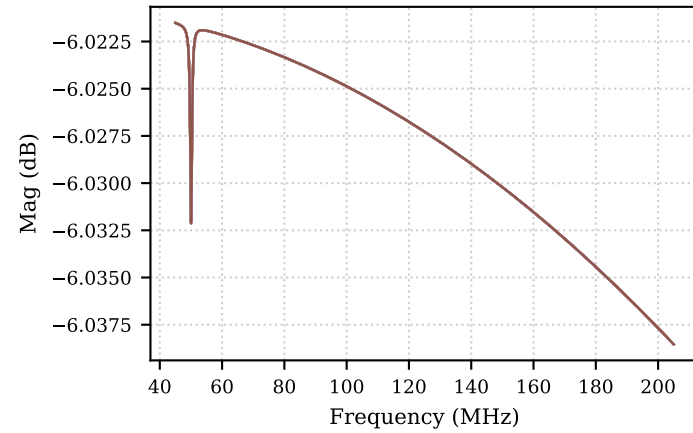
Steps: 88

V(act+,act-) — All curves (88 steps) — Bode

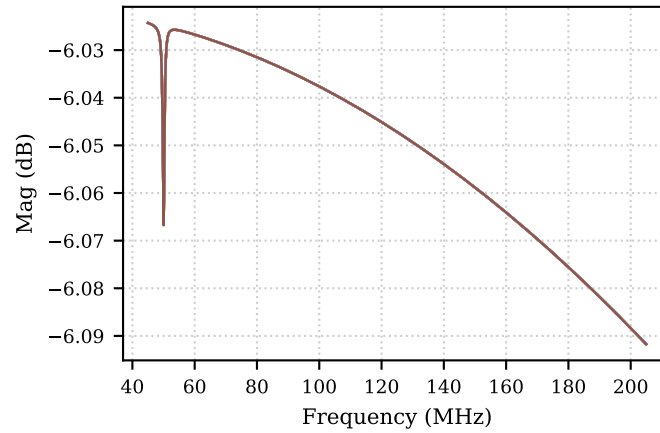


V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)

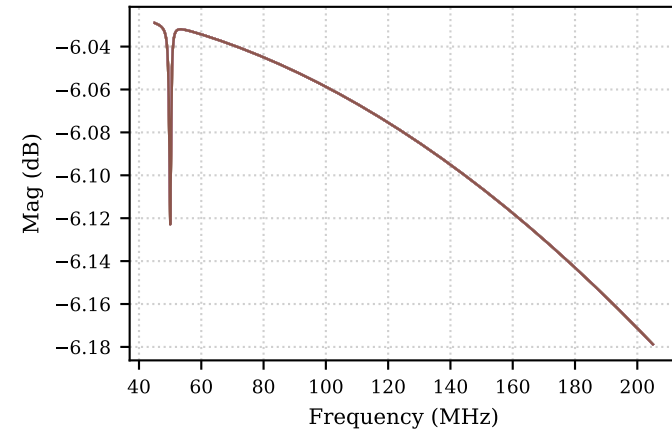
Cp = 100f



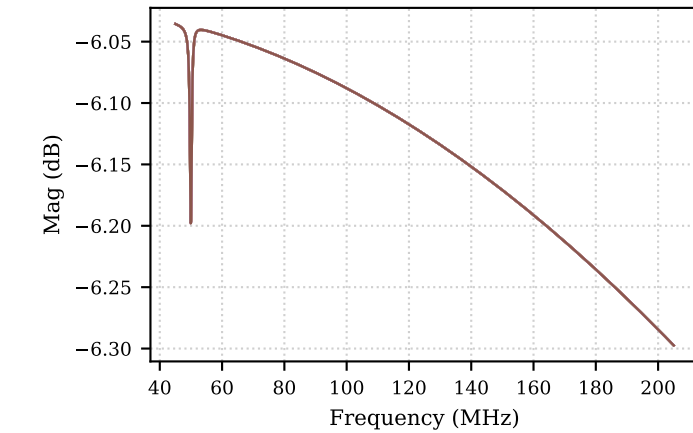
Cp = 200f



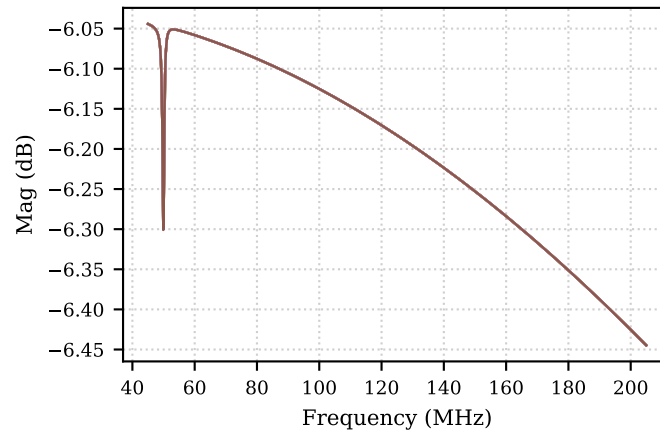
Cp = 300f



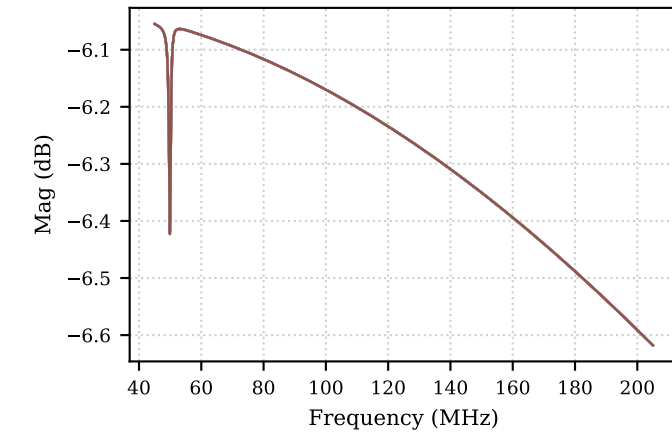
Cp = 400f



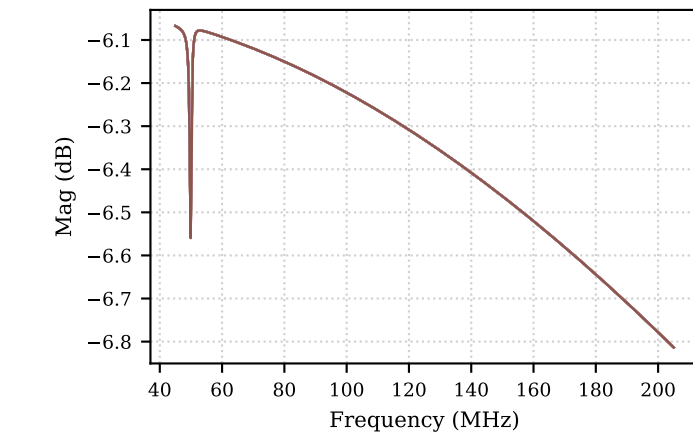
Cp = 500f



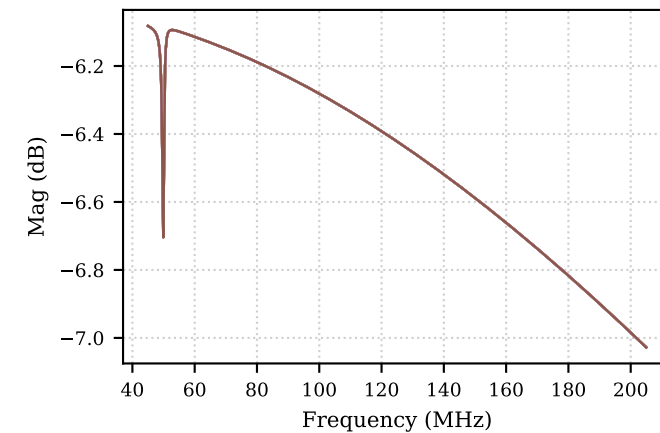
Cp = 600f



Cp = 700f

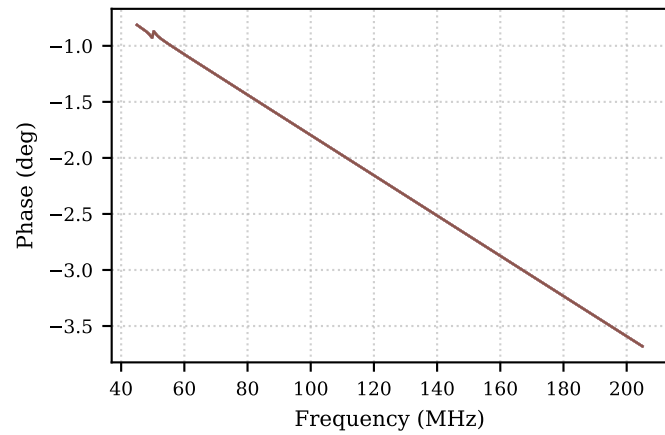


Cp = 800f

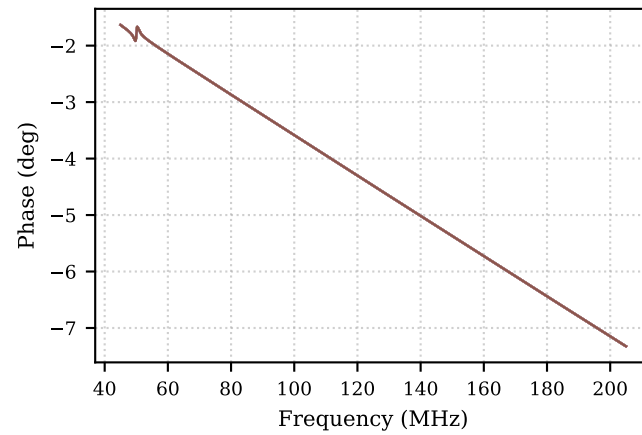


V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)

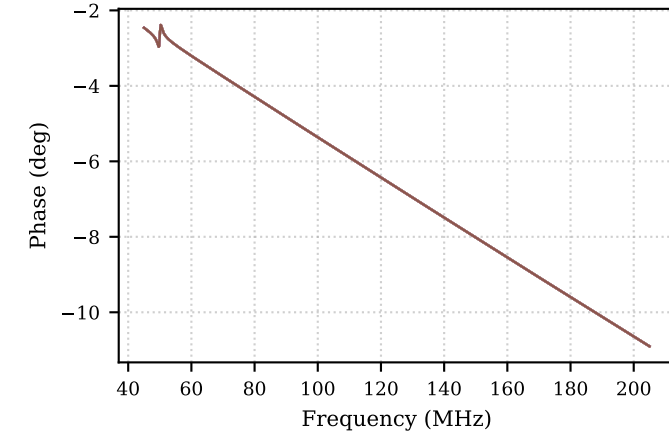
Cp = 100f



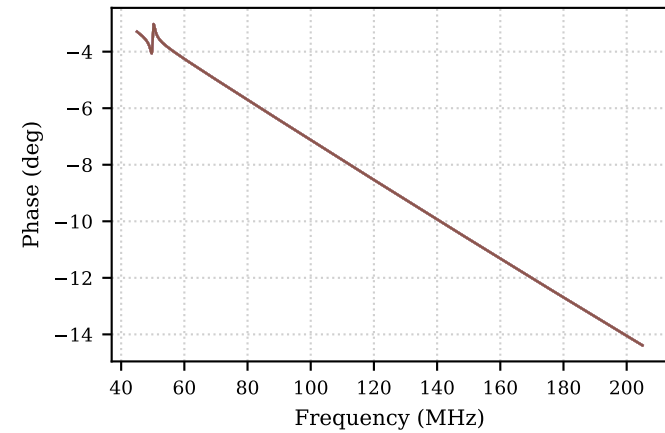
Cp = 200f



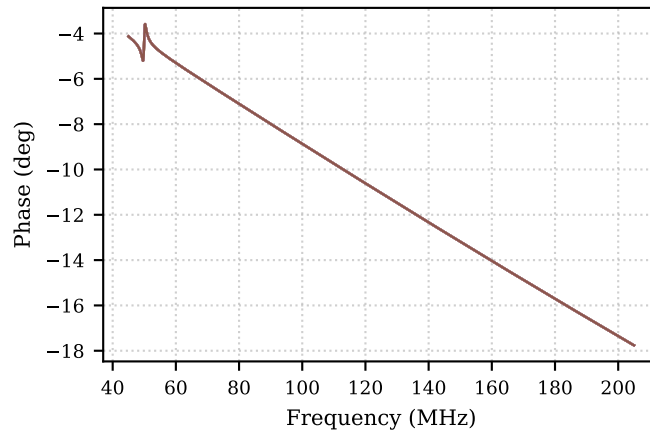
Cp = 300f



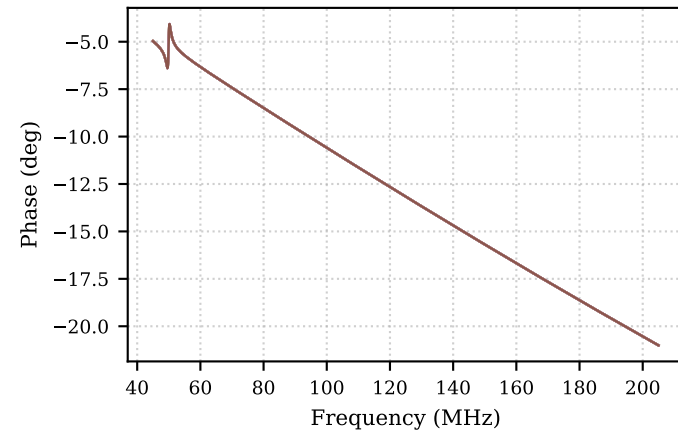
Cp = 400f



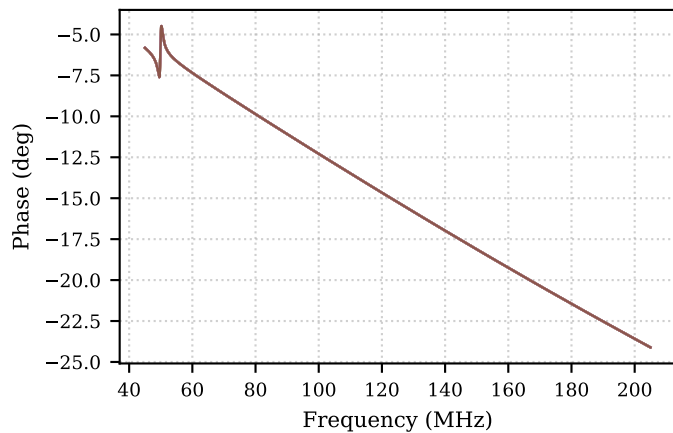
Cp = 500f



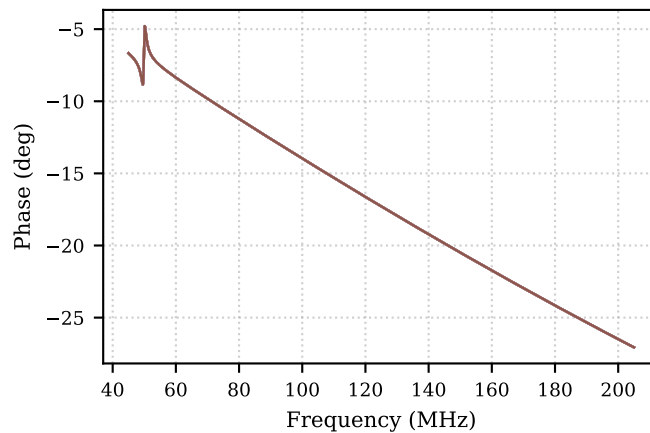
Cp = 600f



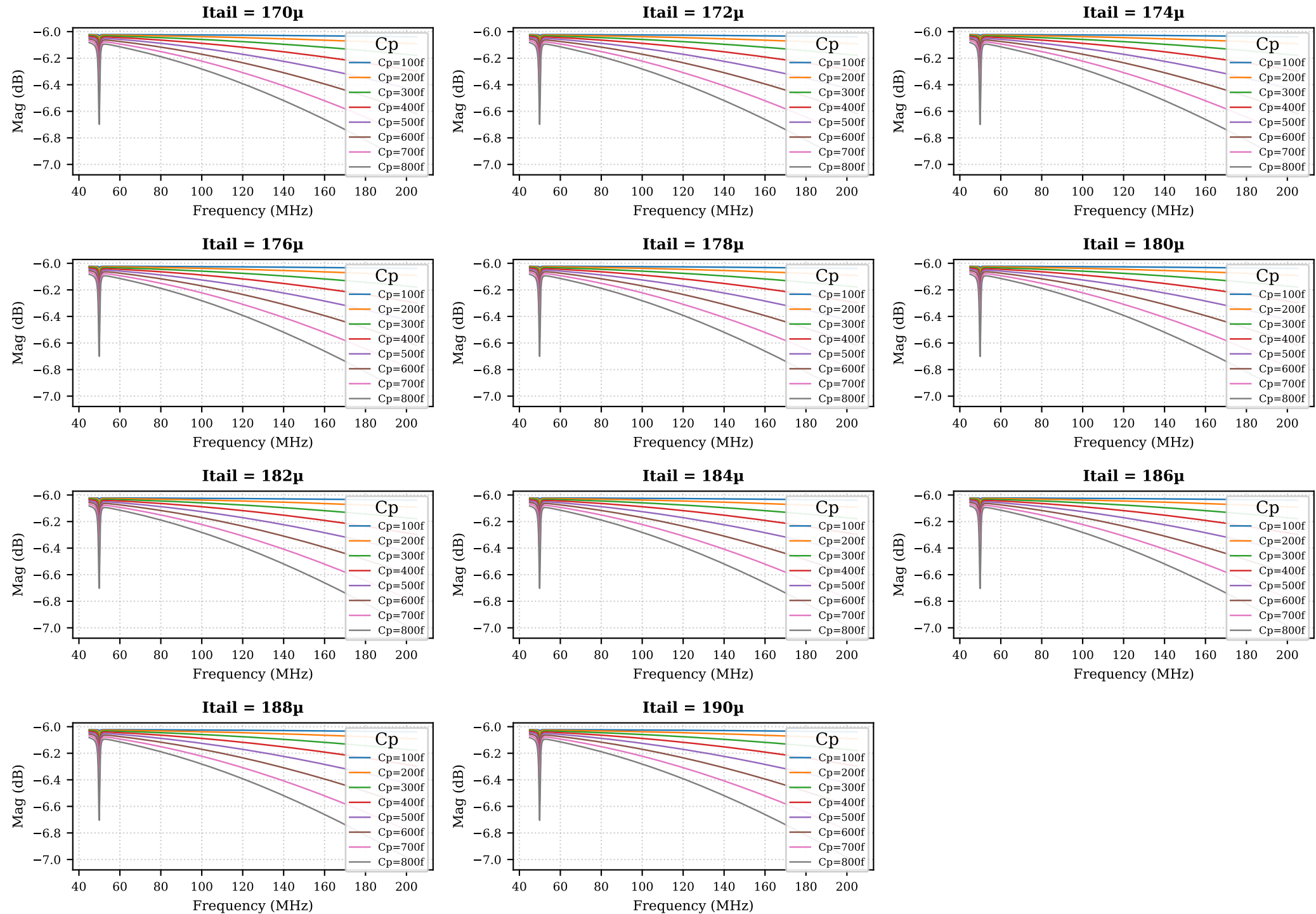
Cp = 700f



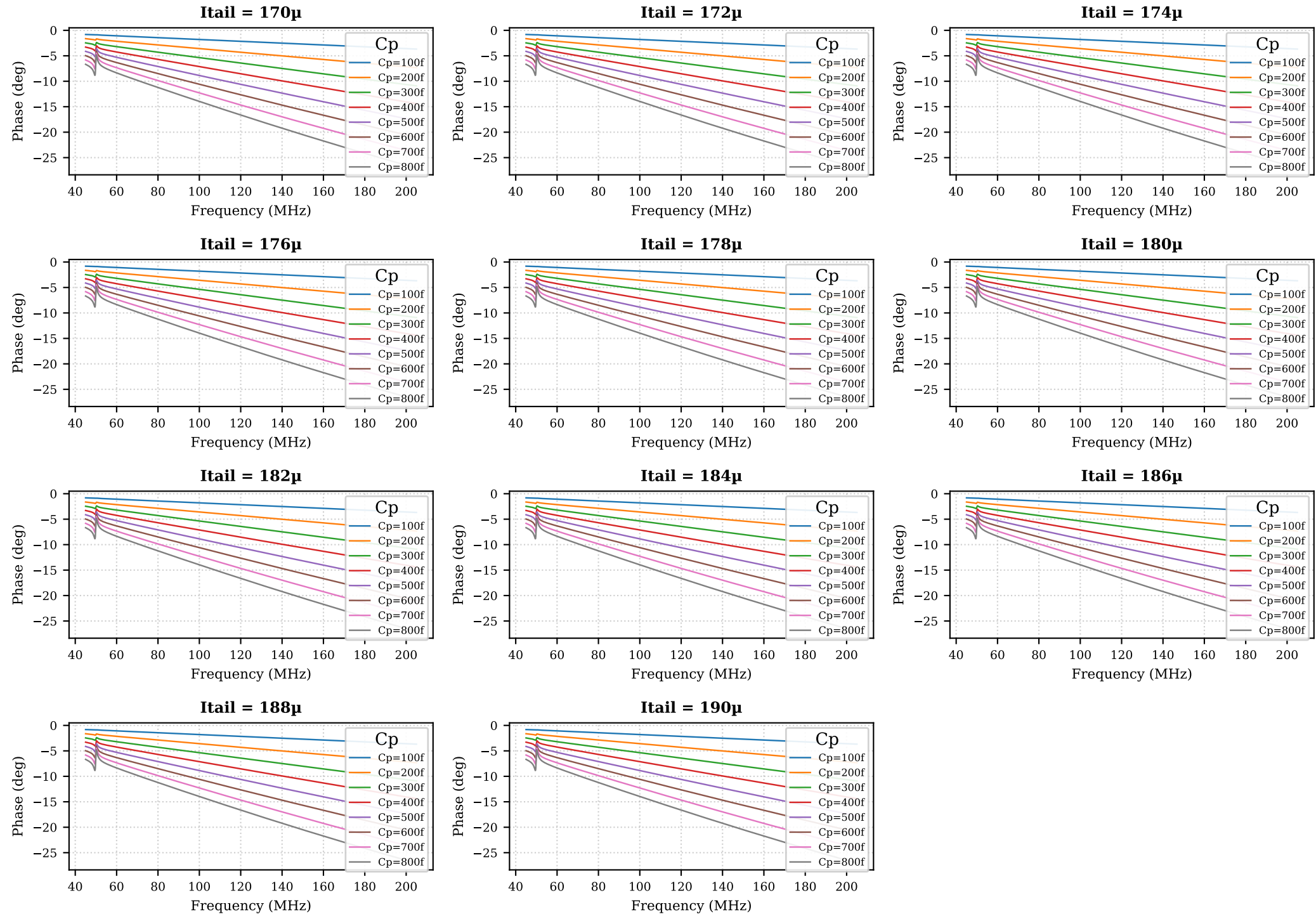
Cp = 800f



$V(\text{act+}, \text{act-})$ — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)



V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)



V(act+,act-) – Quality factor table (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	170μ	205	-6.033	0.005566			notch shallower than 3 d
V(act+,act-)	100f	172μ	205	-6.033	0.005559			notch shallower than 3 d
V(act+,act-)	100f	174μ	205	-6.033	0.005552			notch shallower than 3 d
V(act+,act-)	100f	176μ	205	-6.033	0.005545			notch shallower than 3 d
V(act+,act-)	100f	178μ	205	-6.033	0.005539			notch shallower than 3 d
V(act+,act-)	100f	180μ	205	-6.033	0.005532			notch shallower than 3 d
V(act+,act-)	100f	182μ	205	-6.033	0.005525			notch shallower than 3 d
V(act+,act-)	100f	184μ	205	-6.033	0.005519			notch shallower than 3 d
V(act+,act-)	100f	186μ	205	-6.033	0.005512			notch shallower than 3 d
V(act+,act-)	100f	188μ	205	-6.033	0.005506			notch shallower than 3 d
V(act+,act-)	100f	190μ	205	-6.033	0.005499			notch shallower than 3 d
V(act+,act-)	200f	170μ	205	-6.07	0.02186			notch shallower than 3 d
V(act+,act-)	200f	172μ	205	-6.07	0.02183			notch shallower than 3 d
V(act+,act-)	200f	174μ	205	-6.07	0.0218			notch shallower than 3 d
V(act+,act-)	200f	176μ	205	-6.07	0.02178			notch shallower than 3 d
V(act+,act-)	200f	178μ	205	-6.07	0.02175			notch shallower than 3 d
V(act+,act-)	200f	180μ	205	-6.07	0.02172			notch shallower than 3 d
V(act+,act-)	200f	182μ	205	-6.07	0.0217			notch shallower than 3 d
V(act+,act-)	200f	184μ	205	-6.07	0.02167			notch shallower than 3 d
V(act+,act-)	200f	186μ	205	-6.07	0.02164			notch shallower than 3 d
V(act+,act-)	200f	188μ	205	-6.07	0.02162			notch shallower than 3 d
V(act+,act-)	200f	190μ	205	-6.07	0.02159			notch shallower than 3 d
V(act+,act-)	300f	170μ	205	-6.13	0.04854			notch shallower than 3 d
V(act+,act-)	300f	172μ	205	-6.13	0.04848			notch shallower than 3 d
V(act+,act-)	300f	174μ	205	-6.13	0.04842			notch shallower than 3 d
V(act+,act-)	300f	176μ	205	-6.13	0.04836			notch shallower than 3 d
V(act+,act-)	300f	178μ	205	-6.13	0.0483			notch shallower than 3 d
V(act+,act-)	300f	180μ	205	-6.131	0.04825			notch shallower than 3 d
V(act+,act-)	300f	182μ	205	-6.131	0.04819			notch shallower than 3 d
V(act+,act-)	300f	184μ	205	-6.131	0.04813			notch shallower than 3 d
V(act+,act-)	300f	186μ	205	-6.131	0.04807			notch shallower than 3 d
V(act+,act-)	300f	188μ	205	-6.131	0.04802			notch shallower than 3 d

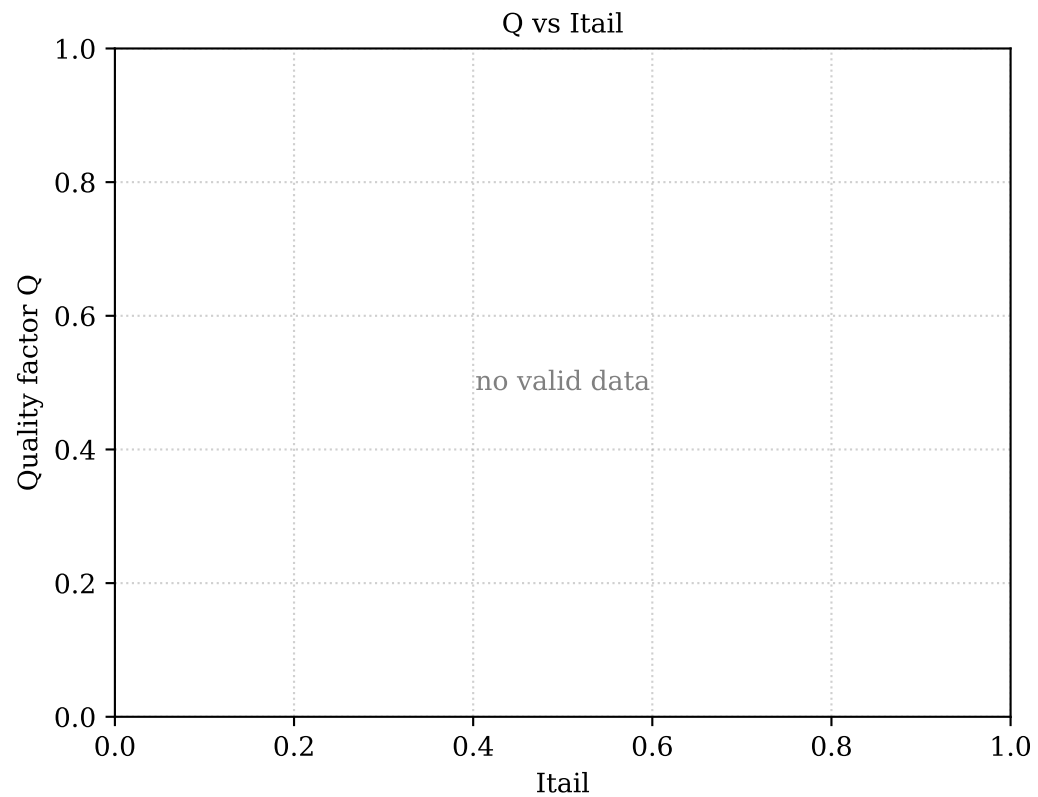
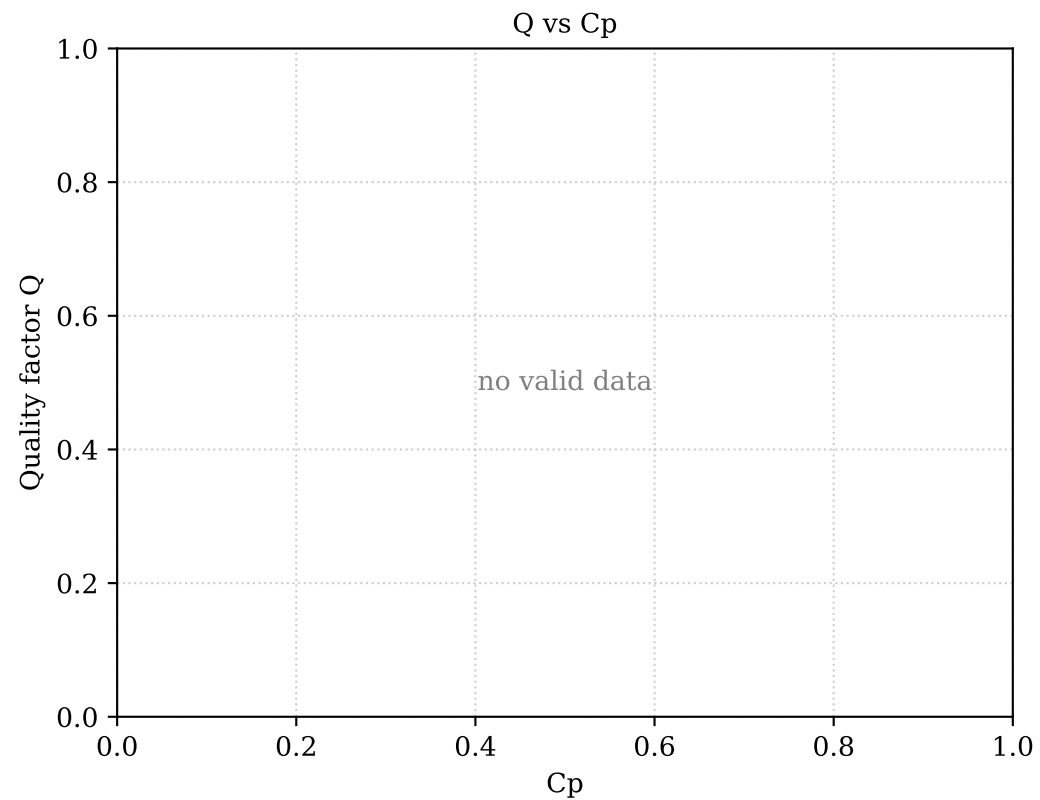
V(act+,act-) — Quality factor table (page 2/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	300f	190μ	205	-6.131	0.04796		notch shallower than 3 d	
V(act+,act-)	400f	170μ	205	-6.212	0.08535		notch shallower than 3 d	
V(act+,act-)	400f	172μ	205	-6.212	0.08526		notch shallower than 3 d	
V(act+,act-)	400f	174μ	205	-6.212	0.08516		notch shallower than 3 d	
V(act+,act-)	400f	176μ	205	-6.212	0.08506		notch shallower than 3 d	
V(act+,act-)	400f	178μ	205	-6.212	0.08497		notch shallower than 3 d	
V(act+,act-)	400f	180μ	205	-6.213	0.08487		notch shallower than 3 d	
V(act+,act-)	400f	182μ	205	-6.213	0.08478		notch shallower than 3 d	
V(act+,act-)	400f	184μ	205	-6.213	0.08468		notch shallower than 3 d	
V(act+,act-)	400f	186μ	205	-6.213	0.08459		notch shallower than 3 d	
V(act+,act-)	400f	188μ	205	-6.213	0.08449		notch shallower than 3 d	
V(act+,act-)	400f	190μ	205	-6.213	0.0844		notch shallower than 3 d	
V(act+,act-)	500f	170μ	205	-6.319	0.1261		notch shallower than 3 d	
V(act+,act-)	500f	172μ	205	-6.319	0.1259		notch shallower than 3 d	
V(act+,act-)	500f	174μ	205	-6.319	0.1258		notch shallower than 3 d	
V(act+,act-)	500f	176μ	205	-6.319	0.1256		notch shallower than 3 d	
V(act+,act-)	500f	178μ	205	-6.319	0.1255		notch shallower than 3 d	
V(act+,act-)	500f	180μ	205	-6.319	0.1253		notch shallower than 3 d	
V(act+,act-)	500f	182μ	205	-6.32	0.1252		notch shallower than 3 d	
V(act+,act-)	500f	184μ	205	-6.32	0.125		notch shallower than 3 d	
V(act+,act-)	500f	186μ	205	-6.32	0.1248		notch shallower than 3 d	
V(act+,act-)	500f	188μ	205	-6.32	0.1247		notch shallower than 3 d	
V(act+,act-)	500f	190μ	205	-6.32	0.1245		notch shallower than 3 d	
V(act+,act-)	600f	170μ	205	-6.445	0.1726		notch shallower than 3 d	
V(act+,act-)	600f	172μ	205	-6.446	0.1724		notch shallower than 3 d	
V(act+,act-)	600f	174μ	205	-6.446	0.1722		notch shallower than 3 d	
V(act+,act-)	600f	176μ	205	-6.446	0.1719		notch shallower than 3 d	
V(act+,act-)	600f	178μ	205	-6.446	0.1717		notch shallower than 3 d	
V(act+,act-)	600f	180μ	205	-6.446	0.1715		notch shallower than 3 d	
V(act+,act-)	600f	182μ	205	-6.447	0.1713		notch shallower than 3 d	
V(act+,act-)	600f	184μ	205	-6.447	0.171		notch shallower than 3 d	
V(act+,act-)	600f	186μ	205	-6.447	0.1708		notch shallower than 3 d	

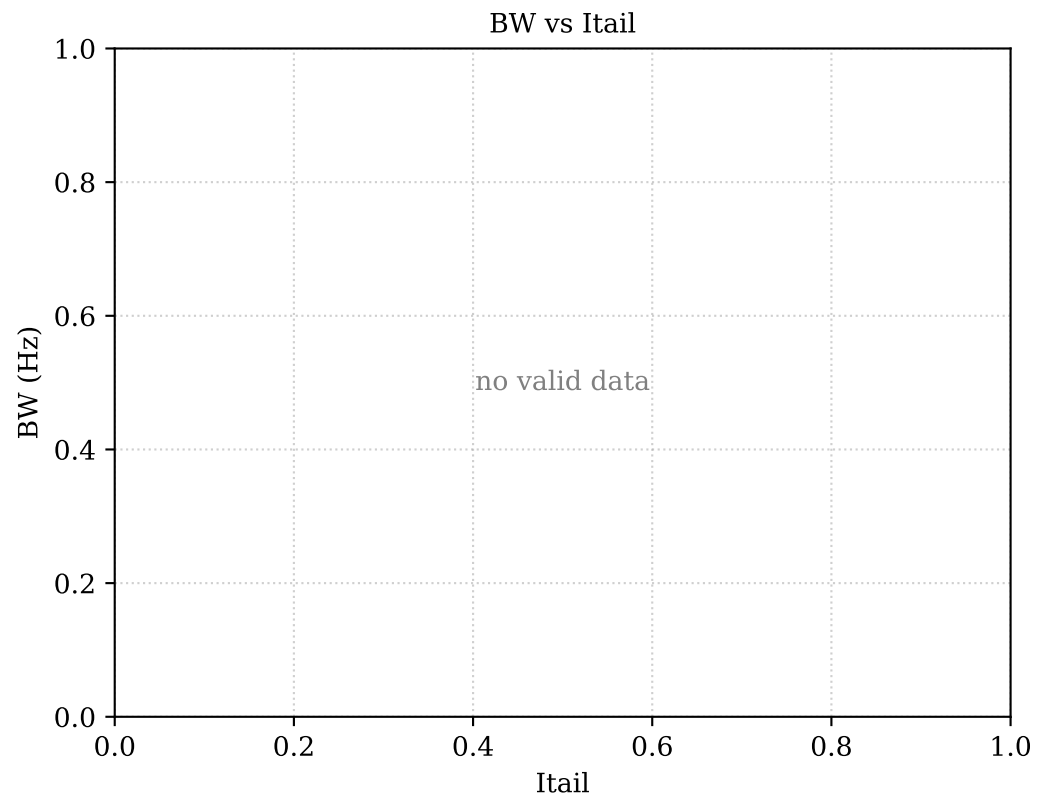
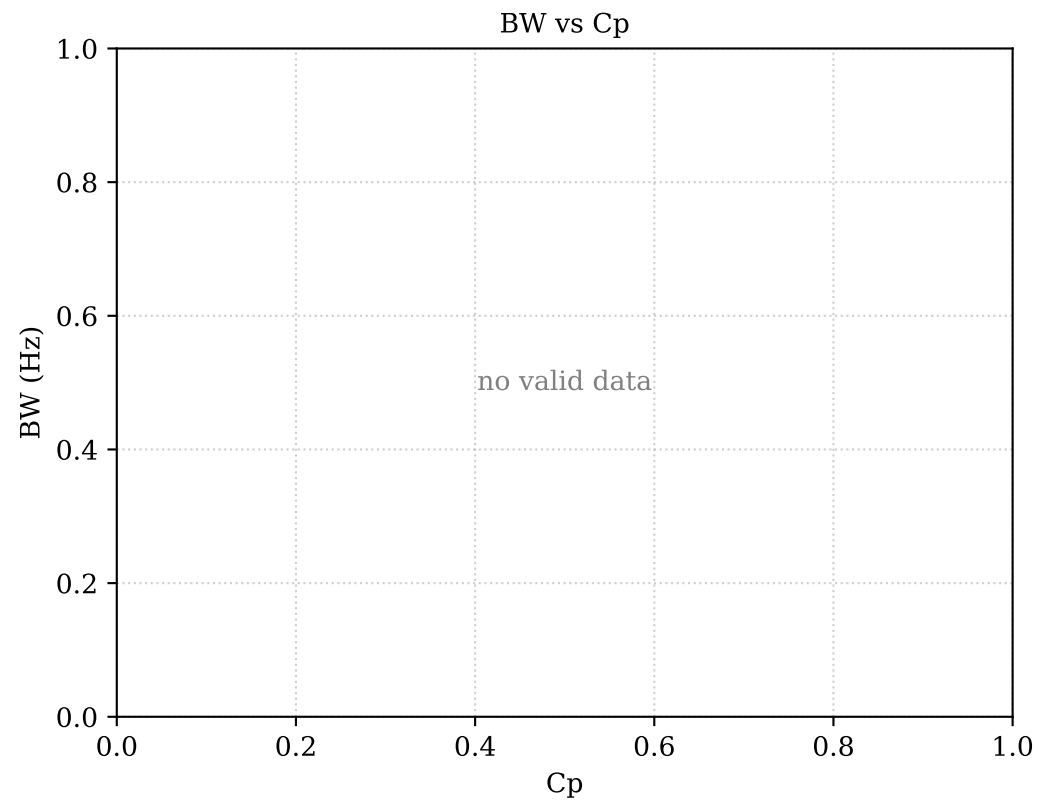
V(act+,act-) — Quality factor table (page 3/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	600f	188μ	205	-6.447	0.1706			notch shallower than 3 d
V(act+,act-)	600f	190μ	205	-6.448	0.1704			notch shallower than 3 d
V(act+,act-)	700f	170μ	205	-6.588	0.225			notch shallower than 3 d
V(act+,act-)	700f	172μ	205	-6.589	0.2247			notch shallower than 3 d
V(act+,act-)	700f	174μ	205	-6.589	0.2244			notch shallower than 3 d
V(act+,act-)	700f	176μ	205	-6.589	0.2241			notch shallower than 3 d
V(act+,act-)	700f	178μ	205	-6.59	0.2238			notch shallower than 3 d
V(act+,act-)	700f	180μ	205	-6.59	0.2235			notch shallower than 3 d
V(act+,act-)	700f	182μ	205	-6.59	0.2232			notch shallower than 3 d
V(act+,act-)	700f	184μ	205	-6.59	0.2229			notch shallower than 3 d
V(act+,act-)	700f	186μ	205	-6.591	0.2226			notch shallower than 3 d
V(act+,act-)	700f	188μ	205	-6.591	0.2223			notch shallower than 3 d
V(act+,act-)	700f	190μ	205	-6.591	0.2221			notch shallower than 3 d
V(act+,act-)	800f	170μ	205	-6.744	0.284			notch shallower than 3 d
V(act+,act-)	800f	172μ	205	-6.744	0.2836			notch shallower than 3 d
V(act+,act-)	800f	174μ	205	-6.745	0.2832			notch shallower than 3 d
V(act+,act-)	800f	176μ	205	-6.745	0.2829			notch shallower than 3 d
V(act+,act-)	800f	178μ	205	-6.745	0.2825			notch shallower than 3 d
V(act+,act-)	800f	180μ	205	-6.746	0.2821			notch shallower than 3 d
V(act+,act-)	800f	182μ	205	-6.746	0.2818			notch shallower than 3 d
V(act+,act-)	800f	184μ	205	-6.746	0.2814			notch shallower than 3 d
V(act+,act-)	800f	186μ	205	-6.747	0.281			notch shallower than 3 d
V(act+,act-)	800f	188μ	205	-6.747	0.2807			notch shallower than 3 d
V(act+,act-)	800f	190μ	205	-6.748	0.2803			notch shallower than 3 d

B. V(act+,act-) — Quality factor vs swept parameters



C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=170μ	Itail=172μ	Itail=174μ	Itail=176μ	Itail=178μ	Itail=180μ	Itail=182μ	Itail=184μ	Itail=186μ	Itail=188μ	Itail=190μ
Cp=100f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=200f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=300f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=400f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=500f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=600f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=700f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=800f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW

E. Combined quality-factor table – all signals (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	170μ	205	-6.033	0.005566			notch shallower than 3 d
V(act+,act-)	100f	172μ	205	-6.033	0.005559			notch shallower than 3 d
V(act+,act-)	100f	174μ	205	-6.033	0.005552			notch shallower than 3 d
V(act+,act-)	100f	176μ	205	-6.033	0.005545			notch shallower than 3 d
V(act+,act-)	100f	178μ	205	-6.033	0.005539			notch shallower than 3 d
V(act+,act-)	100f	180μ	205	-6.033	0.005532			notch shallower than 3 d
V(act+,act-)	100f	182μ	205	-6.033	0.005525			notch shallower than 3 d
V(act+,act-)	100f	184μ	205	-6.033	0.005519			notch shallower than 3 d
V(act+,act-)	100f	186μ	205	-6.033	0.005512			notch shallower than 3 d
V(act+,act-)	100f	188μ	205	-6.033	0.005506			notch shallower than 3 d
V(act+,act-)	100f	190μ	205	-6.033	0.005499			notch shallower than 3 d
V(act+,act-)	200f	170μ	205	-6.07	0.02186			notch shallower than 3 d
V(act+,act-)	200f	172μ	205	-6.07	0.02183			notch shallower than 3 d
V(act+,act-)	200f	174μ	205	-6.07	0.0218			notch shallower than 3 d
V(act+,act-)	200f	176μ	205	-6.07	0.02178			notch shallower than 3 d
V(act+,act-)	200f	178μ	205	-6.07	0.02175			notch shallower than 3 d
V(act+,act-)	200f	180μ	205	-6.07	0.02172			notch shallower than 3 d
V(act+,act-)	200f	182μ	205	-6.07	0.0217			notch shallower than 3 d
V(act+,act-)	200f	184μ	205	-6.07	0.02167			notch shallower than 3 d
V(act+,act-)	200f	186μ	205	-6.07	0.02164			notch shallower than 3 d
V(act+,act-)	200f	188μ	205	-6.07	0.02162			notch shallower than 3 d
V(act+,act-)	200f	190μ	205	-6.07	0.02159			notch shallower than 3 d
V(act+,act-)	300f	170μ	205	-6.13	0.04854			notch shallower than 3 d
V(act+,act-)	300f	172μ	205	-6.13	0.04848			notch shallower than 3 d
V(act+,act-)	300f	174μ	205	-6.13	0.04842			notch shallower than 3 d
V(act+,act-)	300f	176μ	205	-6.13	0.04836			notch shallower than 3 d
V(act+,act-)	300f	178μ	205	-6.13	0.0483			notch shallower than 3 d
V(act+,act-)	300f	180μ	205	-6.131	0.04825			notch shallower than 3 d
V(act+,act-)	300f	182μ	205	-6.131	0.04819			notch shallower than 3 d
V(act+,act-)	300f	184μ	205	-6.131	0.04813			notch shallower than 3 d
V(act+,act-)	300f	186μ	205	-6.131	0.04807			notch shallower than 3 d
V(act+,act-)	300f	188μ	205	-6.131	0.04802			notch shallower than 3 d

E. Combined quality-factor table – all signals (page 2/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	300f	190μ	205	-6.131	0.04796			notch shallower than 3 d
V(act+,act-)	400f	170μ	205	-6.212	0.08535			notch shallower than 3 d
V(act+,act-)	400f	172μ	205	-6.212	0.08526			notch shallower than 3 d
V(act+,act-)	400f	174μ	205	-6.212	0.08516			notch shallower than 3 d
V(act+,act-)	400f	176μ	205	-6.212	0.08506			notch shallower than 3 d
V(act+,act-)	400f	178μ	205	-6.212	0.08497			notch shallower than 3 d
V(act+,act-)	400f	180μ	205	-6.213	0.08487			notch shallower than 3 d
V(act+,act-)	400f	182μ	205	-6.213	0.08478			notch shallower than 3 d
V(act+,act-)	400f	184μ	205	-6.213	0.08468			notch shallower than 3 d
V(act+,act-)	400f	186μ	205	-6.213	0.08459			notch shallower than 3 d
V(act+,act-)	400f	188μ	205	-6.213	0.08449			notch shallower than 3 d
V(act+,act-)	400f	190μ	205	-6.213	0.0844			notch shallower than 3 d
V(act+,act-)	500f	170μ	205	-6.319	0.1261			notch shallower than 3 d
V(act+,act-)	500f	172μ	205	-6.319	0.1259			notch shallower than 3 d
V(act+,act-)	500f	174μ	205	-6.319	0.1258			notch shallower than 3 d
V(act+,act-)	500f	176μ	205	-6.319	0.1256			notch shallower than 3 d
V(act+,act-)	500f	178μ	205	-6.319	0.1255			notch shallower than 3 d
V(act+,act-)	500f	180μ	205	-6.319	0.1253			notch shallower than 3 d
V(act+,act-)	500f	182μ	205	-6.32	0.1252			notch shallower than 3 d
V(act+,act-)	500f	184μ	205	-6.32	0.125			notch shallower than 3 d
V(act+,act-)	500f	186μ	205	-6.32	0.1248			notch shallower than 3 d
V(act+,act-)	500f	188μ	205	-6.32	0.1247			notch shallower than 3 d
V(act+,act-)	500f	190μ	205	-6.32	0.1245			notch shallower than 3 d
V(act+,act-)	600f	170μ	205	-6.445	0.1726			notch shallower than 3 d
V(act+,act-)	600f	172μ	205	-6.446	0.1724			notch shallower than 3 d
V(act+,act-)	600f	174μ	205	-6.446	0.1722			notch shallower than 3 d
V(act+,act-)	600f	176μ	205	-6.446	0.1719			notch shallower than 3 d
V(act+,act-)	600f	178μ	205	-6.446	0.1717			notch shallower than 3 d
V(act+,act-)	600f	180μ	205	-6.446	0.1715			notch shallower than 3 d
V(act+,act-)	600f	182μ	205	-6.447	0.1713			notch shallower than 3 d
V(act+,act-)	600f	184μ	205	-6.447	0.171			notch shallower than 3 d
V(act+,act-)	600f	186μ	205	-6.447	0.1708			notch shallower than 3 d

E. Combined quality-factor table — all signals (page 3/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	600f	188μ	205	-6.447	0.1706			notch shallower than 3 d
V(act+,act-)	600f	190μ	205	-6.448	0.1704			notch shallower than 3 d
V(act+,act-)	700f	170μ	205	-6.588	0.225			notch shallower than 3 d
V(act+,act-)	700f	172μ	205	-6.589	0.2247			notch shallower than 3 d
V(act+,act-)	700f	174μ	205	-6.589	0.2244			notch shallower than 3 d
V(act+,act-)	700f	176μ	205	-6.589	0.2241			notch shallower than 3 d
V(act+,act-)	700f	178μ	205	-6.59	0.2238			notch shallower than 3 d
V(act+,act-)	700f	180μ	205	-6.59	0.2235			notch shallower than 3 d
V(act+,act-)	700f	182μ	205	-6.59	0.2232			notch shallower than 3 d
V(act+,act-)	700f	184μ	205	-6.59	0.2229			notch shallower than 3 d
V(act+,act-)	700f	186μ	205	-6.591	0.2226			notch shallower than 3 d
V(act+,act-)	700f	188μ	205	-6.591	0.2223			notch shallower than 3 d
V(act+,act-)	700f	190μ	205	-6.591	0.2221			notch shallower than 3 d
V(act+,act-)	800f	170μ	205	-6.744	0.284			notch shallower than 3 d
V(act+,act-)	800f	172μ	205	-6.744	0.2836			notch shallower than 3 d
V(act+,act-)	800f	174μ	205	-6.745	0.2832			notch shallower than 3 d
V(act+,act-)	800f	176μ	205	-6.745	0.2829			notch shallower than 3 d
V(act+,act-)	800f	178μ	205	-6.745	0.2825			notch shallower than 3 d
V(act+,act-)	800f	180μ	205	-6.746	0.2821			notch shallower than 3 d
V(act+,act-)	800f	182μ	205	-6.746	0.2818			notch shallower than 3 d
V(act+,act-)	800f	184μ	205	-6.746	0.2814			notch shallower than 3 d
V(act+,act-)	800f	186μ	205	-6.747	0.281			notch shallower than 3 d
V(act+,act-)	800f	188μ	205	-6.747	0.2807			notch shallower than 3 d
V(act+,act-)	800f	190μ	205	-6.748	0.2803			notch shallower than 3 d